Validation Evidence: Hospital Operations & Financial Intelligence

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The dashboard's five headline figures match independent Excel calculations on the raw data, and its slicers change the totals as expected in all seven tests. No mismatches were found.

Method

The raw file encounters.csv was opened in Excel and saved as a separate workbook, so the original stayed untouched. The Excel results were calculated straight from the raw columns with worksheet formulas, using no part of the Power BI model. The Power BI values were then read from a temporary table visual with both slicers cleared and the measures formatted to show exact figures. Each pair was compared in the workbook, and a result counts as a match only when the two agree after rounding to four decimal places.

For the over-24-hour check, duration was calculated in a helper column with ROUND((STOP minus START) x 24, 4). The rounding matters because 1,077 encounters last exactly 24 hours, and without it floating-point noise in Excel's date arithmetic could push some of them above 24 and inflate the count.

Independent checks

Check	Excel formula	Excel result	Power BI value	Result
Total Encounters	COUNTA of the Id column	27,891	27,891	Match
Total Claim Cost	SUM of TOTAL_CLAIM_COST	101,514,375.52	101,514,375.52	Match
Zero Coverage Encounters	COUNTIF of PAYER_COVERAGE equal to 0	13,586	13,586	Match
Zero Coverage %	Zero coverage count divided by total encounters	48.71%	48.71%	Match
Over 24h Encounters	COUNTIF of the rounded hours column above 24	75	75	Match

Supporting check: 1,077 encounters last exactly 24 hours in Excel. They are correctly excluded from the over-24-hour count, which confirms the rounding step worked.

Filter tests

The Year and Encounter Class slicers were set on the Overview page, and the cards were read on the Overview and Encounter Patterns pages. Both slicers were cleared between tests.

#	Slicer selection	Measure read	Expected	Result
1	Year = 2014	Total Encounters	3,885	Pass
2	Year = 2022	Total Encounters	220	Pass
3	Encounter Class = ambulatory	Total Encounters	12,537	Pass
4	Encounter Class = inpatient	Total Encounters	1,135	Pass
5	Encounter Class = inpatient	Over 24h Encounters (Encounter Patterns page)	58	Pass
6	Encounter Class = inpatient	Over 24h % (Encounter Patterns page)	5.11%	Pass
7	Both slicers cleared	Total Encounters	27,891	Pass

Tests 5 and 6 also show that the slicer selection carries across pages, because the Encounter Patterns page changed without any selection being made on it.

Conclusion

All five headline measures agree with independent calculations, and the slicers filter the figures correctly across pages. The dashboard figures can be relied on for the findings in the Executive Summary, subject to the data limitations listed in the Data Quality Assessment.

Evidence files

- Screenshot of the Checks sheet in the Excel validation workbook, showing every row as a match
- Screenshot of the Power BI validation table with both slicers cleared
- Screenshots of the filter tests for Year = 2014, Encounter Class = inpatient on the Overview page, and Encounter Class = inpatient on the Encounter Patterns page
- The Excel validation workbook (encounters_validation.xlsx)